

NEW PRODUCTS

TECHNOLOGY

COMPONENTS & CHIPS

System-in-package for cellular IoT

Nordic Semiconductor is enhancing its nRF91 series with the nRF9151 SiP to cater to the growing cellular IoT market. It provides a complete package of hardware, software, tools, and nRF cloud services, simplifying IoT device development. The compact, pre-certified, integrated device includes an SoC, power management, and an RF front-end. With a 20% smaller footprint than previous models, it is ideal for space-constrained applications. The company highlights that this advancement addresses market demands by streamlining development, reducing power consumption and footprint, and enhancing their cellular IoT portfolio.

Nordic Semiconductor
<https://www.nordicsemi.com>

Smaller MOSFETs with discrete switching

Nexperia has launched MOSFETs to enhance its discrete switching solutions for various applications. The release includes 100V ASFETs for PoE, eFuse, and relay replacement, now in a 60% smaller DFN2020 package, offering improved protection and efficiency. These ASFETs are also suitable for battery management, Wi-Fi hotspots, and CCTV. Additionally, the company introduced

40V NextPowerS3 MOSFETs with enhanced EMC performance, ideal for switching converters and motor controllers.

Nexperia
<https://www.nexperia.com>

Efficient energy-saving MOSFETs

The STPOWER MDmesh DM9 AG series from STMicroelectronics features 600V/650V super-junction MOSFETs are designed for onboard chargers and DC/DC converters in complex



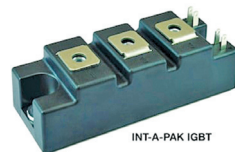
and soft-switching topologies. These silicon-based devices balance low $R_{DS(on)}$ and gate charge for efficient switching and energy savings. With a tighter gate-source threshold voltage spread, they ensure sharper switching, reducing turn-on and turn-off losses. The improved body-diode reverse recovery enhances ruggedness, making them ideal for phase-shift zero-voltage switching topologies. They have AEC-Q101 qualification, low Q_g and $R_{DS(on)}$, fast recovery diode, and high dv/dt and di/dt ruggedness.

STMicroelectronics
<https://www.st.com>

Half-bridge IGBT power modules

Vishay Intertechnology, Inc. has launched five new half-bridge IGBT power modules in the updated INT-A-PAK package, utilising Trench IGBT technology. These modules cater to different design needs, offering either

low $V_{CE(ON)}$ for reduced conduction losses or low E_{off} for minimised switching losses. Ideal for high-current



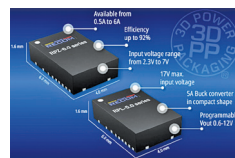
inverter stages in transportation, energy, and industrial applications, they feature

Trench IGBTs with Gen IV FRED Pt anti-parallel diodes for soft reverse recovery. The INT-A-PAK package is compatible with the 34mm industry standard, allowing easy replacement.

Vishay Intertechnology
<https://www.vishay.com>

DC/DC converters with integrated inductors

RECOM's new RPZ series of step-down non-isolated DC/DC converters with integrated inductors are designed for compact board spaces.



They feature flip-chip technology, up to 92% efficiency, and can operate at

high ambient temperatures, such as 125°C for the RPZ-1.0 with derating. All models are protected against short circuits, over-current, over-temperature, and input under-voltage. Typical applications include imaging systems, distributed power architectures, optical modules, and portable battery-powered equipment in telecom and industrial settings.

RECOM power
<https://recom-power.com>

RF fixed attenuators with 3.5mm connectors

Pasternack has launched its new RF fixed attenuators with 3.5mm connec-